

SYNOPSIS



The Marine Accident Investigation Branch (MAIB) was notified of the accident on 6 January 2002 and an investigation began that day.

The high-speed craft *Diamant* and the ro-ro passenger ferry *Northern Merchant* collided in the Dover Strait in poor visibility.

Diamant was en-route from Oostende to Dover with 148 passengers and crew on board. *Northern Merchant* had just departed Dover for Dunkerque with 102 passengers and crew on board.

As both vessels approached each other with a CPA of 3 cables, 3 miles south-east of Dover, *Diamant's* speed was 29 knots, *Northern Merchant's* was 21 knots. The bridge team on *Diamant* then assumed, incorrectly, a green - green situation and maintained course and speed.

On board *Northern Merchant*, the bridge team fully expected *Diamant* to keep clear, because of a perceived unwritten rule that high-speed craft will keep clear of all other vessels in all scenarios. However, as the distance between the vessels decreased to 6 to 7 cables, they realised this might not be the case and then altered course to starboard by 7° to 10° and then applied 20° of helm. At the same time, *Northern Merchant's* echo began to arc on *Diamant's* radar. The master of *Diamant*, assuming the danger to be on his starboard side, altered course to port. The result was that the vessels collided.

As a result of the collision, *Diamant* suffered substantial prow and starboard side wave piercer damage. *Northern Merchant* suffered slight damage to her port side shell plating. There were no injuries.

This accident has raised three important safety issues relating, firstly, to the perceived unwritten rule, secondly, how operators should determine a safe speed and close quarter situation in restricted visibility and, thirdly, the extent to which reliance can be placed on radar for detection in restricted visibility.

As a result of the issues, appropriate recommendations have been made to the Maritime and Coastguard Agency and the operator of each vessel.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *DIAMANT/NORTHERN MERCHANT* AND ACCIDENT

Vessel details

Name	:	<i>Diamant</i> (Figure 1)
Registered owner	:	Hoverspeed Ltd, Dover, Kent
Operator	:	Sea Containers Ltd
Port of registry	:	Luxembourg
Flag	:	Luxembourg
Type	:	Passenger/ro-ro cargo ferry, twin hull
Built	:	1996 Incat Australia Pty Ltd, Hobart, Tasmania
Classification society	:	Det Norske Veritas
Construction	:	Aluminium alloy
Length overall	:	81.15m
Gross tonnage	:	4,305
Engine power and type	:	22000kW 4x Ruston Paxman Diesels
Service speed	:	40 knots
Other relevant info	:	4 x water jet propulsion units

Accident details

Persons on board	:	148
Injuries	:	None
Damage	:	Substantial prow and starboard side wave piercer damage
Location of accident	:	51° 06.3' N 001° 26.1' E
Date and Time	:	0951 (UTC+1) on 6 January 2002

Vessel details

Name	:	<i>Northern Merchant</i> (Figure 2)
Registered owner	:	Genargo Ltd, Puttenham, UK
Manager	:	V Ships Ltd Glasgow
Port of registry	:	Dover
Flag	:	UK
Type	:	Passenger/ro-ro cargo ferry
Built	:	2000 Astilleros Espanoles S.A., Spain
Classification society	:	Lloyd's Register of Shipping
Construction	:	Steel
Length overall	:	179.93m
Gross tonnage	:	22,152
Engine power and type	:	23,760 kW 4x Wartsila diesels
Service speed	:	23 knots
Other relevant info	:	Quadruple screw

Accident details

Persons on board	:	103
Injuries	:	None
Damage	:	Slight port side shell plate damage
Location of accident	:	51° 06.3' N 001° 26.1' E
Date and Time	:	0951 (UTC+1) on 6 January 2002

1.3 BACKGROUND

1.3.1 Vessels

Diamant was purchased by her current owner, Hoverspeed Ltd, a subsidiary company of Sea Containers Ltd, on 6 March 1998. Before this date, the builder Incat, delivered the vessel (Incat 041) to Denmark to operate between the ports of Arhuss and Calanberg.

Shortly after, on 1 March 1997, she was purchased by Sally Hollyman Ltd and named *Hollyman Diamant* for operation on the Ramsgate - Oostende route. During this time, Condor Marine Services was responsible for marine and technical services. This role was taken over by Sea Containers Ltd following her sale to Hoverspeed.

In addition to *Diamant*, Sea Containers Ltd operate five 74m and one other 81m Incat Sea-Cats, on routes between the UK, Eire, France and Denmark. In addition, it also operates three Superseacats and two conventional ro-ro passenger class II vessels.

At the time of the accident, *Diamant* was engaged on a regular cross-Channel route between Dover and Oostende, a crossing which normally took approximately 2¼ hours. During a normal day she was scheduled to cross the English Channel four times, leaving Oostende on the first sailing of the day at 0845 and departing Dover at 1845 on the last sailing.

Northern Merchant, managed by V Ships, which was responsible for marine and technical services, was, and had been since being built, on a time charter to Norfolk Line which engaged her in a regular cross-Channel service between Dover and Dunkerque. The service was complemented by her sister vessel *Midnight Merchant*. Each vessel was scheduled to cross the English Channel six times a day with each crossing taking approximately 2 hours.

V Ships also operated another six ro-ro/passenger ferries. Two, *Brave Merchant* and *Dawn Merchant*, sister vessels to *Northern Merchant* and *Midnight Merchant*, on the Liverpool to Dublin route; *Lagan Viking* and *Mersey Viking*, on the Liverpool to Belfast route; and *Scirocco* and *Mistral* in the Mediterranean on the Almeria to Nador route.

1.3.2 Dover Strait

The Dover Strait and its approaches is one of the busiest waterways in the world, and poses particular safety concerns because of the density of traffic and the proximity of navigational hazards. In 1977, the traffic separation scheme (TSS) in the Dover Strait and adjacent waters, became compulsory.

The area experiences reduced visibility throughout the year, although it is rare in July and August. During January and February fog is experienced on about 4 days each month.

1.5 ENVIRONMENTAL CONDITIONS

At the time of the accident the weather conditions were a north-westerly wind, force 2, with slight seas. The visibility was poor to very poor in areas of dense fog. The tide was ebbing in a direction of 219° at a rate of 1.4 knots, and the times of high and low water at Dover were 0400 and 1125 respectively.

1.6 NARRATIVE OF EVENTS (ALL TIMES ARE UTC +1) (ALL COURSES ARE TRUE)

At 0743, after boarding passengers and loading vehicles, *Diamant* left the port of Oostende en-route for Dover. The visibility was approximately 300 to 500 metres.

The master, mate, chief engineer and a lookout were on the bridge. They had all joined the vessel that morning, after spending the night at home. As soon as *Diamant* passed the piers, at 0747, the automatic fog signal was started and another lookout was called. A lookout was then posted on either side of the bridge.

The master, who had the conduct of the navigation, was sitting in the left-hand seat, with his radar (3cm) offset on the 6-mile range. The chief officer was sitting in the right-hand seat, the navigator's seat, with his radar (10cm) also offset on the 6-mile range. Both radars were displaying water-based information. It was normal practice to operate the radars in Fott, changing to relative motion for collision avoidance. During the crossing of the TSS, the range of both radars was increased to 12 miles, all targets acquired, then reduced to the 6-mile range once clear of the TSS.

There was very little traffic as *Diamant* began crossing the Channel.

Shortly after leaving Oostende, all four engines were increased to 930 rpm giving a speed of 32 knots. However, at 0755, the port inner engine tripped because an oil mist detector alarmed. With only three engines in operation, the speed was reduced to 24 knots.

At 0825, the problem with the port inner engine was rectified and 25 minutes later a speed of 32 knots was once again attained. The ETA at Dover was revised from 0945 to 1000.

At 0830 in Dover harbour, *Northern Merchant* began loading freight and boarding passengers. The operation was completed by 0920. After loading, the chief officer, who was supervising on deck, made his way to the bridge where the master, who had been there for some time, was making preparations for departure. The visibility in Dover at that time was approximately 150 to 200 metres.

At 0930 *Northern Merchant* let go. On doing so, her automatic fog signal was activated. The master, chief officer, and also an AB/helmsman, who had made his way there after helping let go forward, were on the bridge. The master had the conduct of the navigation while the chief officer monitored the starboard radar, which was set to the 0.25 mile range. The port radar was set to the 6-mile range.

At the same time, *Diamant*, still at 32 knots, having crossed the TSS, was passing CS4 Buoy on a course of 250° for the approach to Dover's western entrance. Both the master and the chief officer had heard *Northern Merchant* request permission, on VHF radio channel 74, from Dover harbour to depart.

After letting go, *Northern Merchant* maintained her position in harbour to allow *Cezanne*, another outgoing ferry, to depart. *Northern Merchant* then followed *Cezanne* through the breakwater at a distance of approximately 3 cables.

At 0940, once clear of the breakwater, on a course of 120°, the chief officer increased the range of his radar to 6 miles and the master then ordered a course of 105° and began to increase speed. However, as he was doing so he became aware that *Calais*, another ferry, was inward bound. He then ordered an alteration of course to 115° to pass to the north of *Cezanne*, and reduced speed to 12 knots. At the same time, the bridge team on *Diamant* were made aware that *Hoverspeed Great Britain*, another HSC, was going to be late in departing the berth; consequently, they reduced speed to 29 knots to allow more time. The time was 0943.

As soon as *Cezanne* and *Northern Merchant* were clear of Dover harbour, both the master and chief officer on board *Diamant* began plotting them on the vessel's ARPA's, from which they interpreted the following information relating to the target they considered to be *Northern Merchant*: course 085°, CPA 3.0 cables. The other target was passing clear to the south on a course of 120°. Course and speed were maintained.

At 0945, as *Diamant* passed South Goodwin light vessel, the visibility began to deteriorate. Both radars' ranges were then reduced from 6-miles to 3-miles with the closing target being monitored constantly.

At approximately the same time, after passing clear to the north of *Cezanne*, the master on board *Northern Merchant* ordered a resumption of course to 105° and increased speed to 20 knots. He then engaged the autopilot. He also heard on VHF radio, *Diamant*, requesting permission to enter Dover harbour. The master was monitoring the starboard radar and the chief officer the port radar. Both radars were set to the 6-mile range and the information displayed was ground-based, relative vectors. Shortly after settling on the resumed course, the master informed the chief officer that "a situation" was developing with respect to a

target near the South Goodwin light vessel. From the ARPA radars monitored by the master and chief officer, they interpreted the following information: course approximately 230°, range 2 miles, speed 32 knots, CPA 2.0 cables. The master then instructed the helmsman to return to the wheel. Course was then altered 7° to 10° to starboard.

At 0949, after requesting permission for entry to Dover harbour, the bridge team on board *Diamant* interpreted, again from the vessel's ARPAs, the following information on the closing target, which they considered to be a green-green situation: range 6 to 7 cables, bearing 023° relative, CPA 2.0 cables. They instructed the two lookouts to keep a sharp visual and listening watch. The chief officer then started giving the master constant range readouts of the target. Shortly after this the CPA increased to 3 cables.

On board *Northern Merchant*, at approximately 0951, with no improvement in the situation, the master, in an attempt to avoid a collision, ordered 20° starboard helm and began rapidly sounding the fog signal. The chief officer and the second officer then went to the port side of the bridge and joined a fitter who was working there. They all heard the engines of an approaching vessel getting louder.

Back on *Diamant*, when the range reduced to 4 to 5 cables, the echo of the target began to arc through about 180° from slightly to port of *Diamant's* heading marker around her starboard side to abaft the beam on the radar screen. The master began to alter course to port, initially by applying 5° helm, and then hard over when *Northern Merchant's* port side became visible out of the fog, at a distance of approximately 50 to 150 metres. He then reversed the port engines in an attempt to increase the rate of turn. Realising a collision was unavoidable, the chief officer warned the passengers of an imminent collision and instructed them to sit down. Shortly after, at approximately 0952, *Diamant* collided with the port side of *Northern Merchant* at a point slightly aft of amidships, her prow making contact first, followed by her starboard wave piercer. Fortunately, as both vessels were turning away from each other, the collision was more a glancing blow than "square-on" contact.

Immediately after the collision, the chief officer called *Northern Merchant* on VHF radio channel 74 to enquire as to her status and to offer any assistance. The chief officer on *Northern Merchant* replied that there appeared to be some minor damage.

After reporting the situation to Dover coastguard, the chief officer on *Northern Merchant* instructed the crew to check for any further damage. Meanwhile, on board *Diamant*, the crew and engineers also checked for damage and sounded round. The cabin crew were instructed to report any passenger injuries. There were none.

After soundings and checks were made, it was reported that there was substantial damage to No.1 void space in the starboard hull (**Figure 6**) with ingress of water, but the bilge pumps were coping. Damage to some cars was also reported. Ten minutes after the collision, the master made a further announcement to the passengers to inform them the vessel would be continuing to Dover at slow speed.

Northern Merchant then contacted *Diamant* to inform the master she had sustained only minor damage, and if no assistance was required, she would continue her passage to Dunkerque.

At 1005, *Diamant* obtained permission to enter Dover harbour, after which the master informed the passengers that the vessel was fully manoeuvrable and they would shortly be arriving in port.

At 1026, *Diamant* was securely moored alongside in Dover. Soon after, the passengers were disembarked.

1.7 HIGH-SPEED CRAFT

1.7.1 Manoeuvring

Unlike conventional ships, high-speed craft have inherently different manoeuvring characteristics. The catamaran has a significantly larger deck area when compared with a mono-hull of the same displacement (usually in the order of 50% greater). They also exhibit high transverse stability and high speed capability where the hulls are designed in a semi-planing form. Low draught and lightness of construction are also advantages of the catamaran.

Propulsion is normally provided by diesel engines (two in each engine room) coupled to water jets giving in excess of approximately 35 tonnes thrust each at maximum acceleration. Thrust vectoring and steering is provided on all jets.

On average, at operating speeds of between 30 and 40 knots, a rate of turn of 130° - 145° per minute can be achieved, with the completion of a full circle in less than 500 metres laterally and transversely. In addition, crash stopping distances are usually in the order of no more than 4 to 5 times the length of the vessel. *Diamant's* crash stopping distance was 463 metres, carried out during sea trials before delivery.

Another feature of this type of vessel is the capability of “momenting”; a manoeuvre carried out with both sets of water jets directed inwards towards the centre line of the vessel or outwards away from it. When employed, the vessel can be turned through 360° in the vessel's own length. However, this is only practical at low speeds due to the danger of substantial damage to the manoeuvring buckets at high speeds.

Even though high-speed craft are deemed to be highly manoeuvrable, they are restricted to certain operational speeds. All high-speed craft have a range of critical speed, ie speeds which should be avoided, because of either potential damage/vibration to the vessel, or the wave wash generated, which can be dangerous to other vessels and shorelines.

Diamant's critical speed was 14.5 – 25 knots.

1.7.2 History of high-speed vessels in UK waters

Hovercraft were first used on a regular commercial service in the UK on the Solent between Southsea and Ryde in the mid-1960s. This is the last commercial hovercraft service still in operation in the UK.

In the late 1960s, commercial hovercraft began operating across the Dover Strait. In 1969, hydrofoils were used to cross the Solent between Southampton and Cowes.

From the beginning of the 1990s, high-speed catamarans and mono-hulls have been used on a variety of routes to and from UK ports. In 1996, Stena Line introduced the first of three high-speed ferries known as HSS. These ferries are the largest high-speed craft operating anywhere in the world, and can carry up to 1500 passengers and 375 cars, at service speeds in excess of 40 knots.

1.7.3 Previous accidents involving high-speed craft

Among international accidents involving high-speed vessels are:

- The *Apollo Jet* grounding in Hong Kong harbour on 15 December 1989, resulting in four dead and seven seriously injured.
- The collision off the coast of British Columbia, Canada on 6 February 1992, between the ferry *Queen of Saanich* and the passenger catamaran ferry *Royal Vancouver*, in which 23 were injured.
- The *Sleipner* grounding off the Norwegian coast on 26 November 1999, in which 15 were killed along with one missing.

In the UK, the MAIB has investigated the following accidents involving high-speed vessels:

- The grounding of the catamaran *St Malo* off Corbiere Point, Jersey on 17 April 1995 in which 55 passengers were injured.
- The hovercraft *Princess Margaret* struck a boundary wall while berthing in Dover in September 1997, causing damage but no injuries.
- An angler on the small boat *Purdy* was washed overboard and drowned in July 1999 after his vessel was swamped by a wave from the HSS *Stena Discovery*.

1.9 COLLISION REGULATIONS

1.9.1 Rule 2

The International Regulations for Preventing Collisions at Sea, Rule 2, Responsibility, states:

- (a) *Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.*
- (b) *In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessel involved, which may make a departure from these Rules to avoid immediate danger.*

1.9.2 Rule 5

Rule 5, Look-out states:

Every vessel shall at all times maintain a proper lookout by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

1.9.3 Rule 6

Rule 6, Safe Speed states:

Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions.

In determining a safe speed the following factors shall be among those taken into account:

- (a) *By all vessels:*
 - (i) *the state of the visibility;*
 - (ii) *the traffic density;*
 - (iii) *the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;*
 - (iv) *at night the presence of background light such as from shore lights or from back scatter of her own lights;*

- (v) *the state of the wind, sea and current, and the proximity of navigational hazards;*
- (vi) *the draught in relation to the depth of water.*
- (b) *Additionally, by vessels with operational radar:*
 - (i) *the characteristics, efficiency and limitations of the radar equipment;*
 - (ii) *any constraints imposed by the radar range scale in use;*
 - (iii) *the effect on radar detection of the sea state, weather and other sources of interference;*
 - (iv) *the possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range;*
 - (v) *the number, location and movement of vessels detected by radar;*
 - (vi) *the more exact assessment of the visibility that may be possible when radar aids used to determine the range of vessels or other objects in the vicinity.*

1.9.4 Rule 7

Risk of Collision states:

- (a) *Every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt such risk shall be deemed to exist.*
- (b) *Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.*
- (c) *Assumptions shall not be made on the basis of scanty information, especially scanty radar information.*
- (d) *In determining if risk of collision exists the following considerations shall be among those taken into account:*
 - (i) *such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change;*
 - (ii) *such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.*

1.9.5 Rule 8

Rule 8, Action to Avoid Collision states:

- (a) *Any action taken to avoid collision shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.*
- (b) *Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.*
- (c) *If there is sufficient sea room, an alteration of course alone may be the most effective action to avoid a close quarters situation provided that it is made in good time, is substantial and does not result in another close quarters situation.*
- (d) *Action taken to avoid collision with another vessel shall be such as to result in passing at a safe distance. The effectiveness of the action shall be carefully checked until the other vessel is finally past and clear.*

1.9.6 Rule 19

Rule 19, Conduct of Vessels in Restricted Visibility, states:

- (a) *This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.*
- (b) *Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate manoeuvre.*
- (c) *Every vessel shall have due regard to the prevailing circumstances and conditions of restricted visibility when complying with the Rules of section 1 of this part.*
- (d) *A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:*
 - (i) *an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;*
 - (ii) *an alteration of course towards a vessel abeam or abaft the beam.*

- (e) *Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over.*

1.10 NAVIGATION IN FOG

Advice concerning navigation in fog is contained in *Marine Guidance Note MGN 202 (M&F)*, published by the MCA and entitled *Navigation in Fog*, (**Annex 1**) which states, in part:

The Maritime and Coastguard Agency (MCA) is concerned that a number of casualties to ships have resulted from serious disregard for the basic principles of good seamanship and prudent navigation in bad visibility. Sensible use of radar and other aids to navigation greatly assists the conduct of ships in fog, but these aids have not reduced the need to comply fully with the Collision Regulations: to proceed at a safe speed, pay special attention to good watch-keeping, and navigate with proper caution.

It gives a brief outline of three casualties in fog, then states:

None of the casualties described led to loss of life, but clearly this was only due to good fortune. In all cases those responsible for the ship's navigation sacrificed seamen for expediency. They failed to recognise the limitations of aids to navigation; or to follow the requirements of the Collision Regulations and the advice of Marine Notices. It is worth stressing that the ships involved were all well-equipped vessels in the charge of men with sound qualifications; it was not skill or experience that was lacking, but the proper seamanlike approach to the situation.

Whatever the pressure on masters to make a quick passage or to meet the wishes of owners, operators, charters or port operators, it does not justify ships and those on board them being put unnecessarily at risk.

The document also stresses the responsibilities of owners; it is the duty of the company to take all reasonable steps to ensure that the ship is operated in a safe manner. In this regard, it states:

The company must have established and implemented an effective Safety Management System which includes procedures to ensure safe operation of ships as well as reporting accidents and non-conformities.

1.11 COMPANY PROCEDURES/RISK ASSESSMENT

Both *Northern Merchant* and *Diamant* had been issued with a safety management certificate and had in place, in compliance with ISM requirements, guidance and instructions to masters and crew in the form of operating manuals.

In the case of *Northern Merchant*, her *Ship Operating Manual (SOM)* (**Annex 2**) under *Part Four – The Ship at Sea - Special Requirements in Bad Weather and Fog* states in part:

Prevention of Collision

We would bring to the master's attention the need and requirement for deck officers to review the Rules for Preventing Collisions at Sea 1972 in their entirety but with specific reference to the following:

Rule 2 Responsibility

Rule 6 Safe Speed

Rule 19 Conduct of Vessels in Restricted Visibility

We would stress that it is the bridge watchkeeper's prime duty to ensure that these Rules are complied with at all times on the vessel and that proper application of these rules will take precedence over the commercial pressure or requirements of the vessel.

Fog and Restricted Visibility

Prior to entering fog or areas of reduced visibility the main engines must be slowed and precautions carried out as per the ICS Bridge Procedures Guide.

In the case of *Diamant*, her *Route Operating Manual* (**Annex 3**), under section 3.6.4 *Navigation – Planning and Execution* states in part:

Commercial considerations must not take precedence over good seamanship. In poor visibility or if there is any doubt about the craft's position, or in the presence of other traffic/obstacles, speed should be reduced to that which is considered safe in the circumstances.

3.7.5 Collision Avoidance states:

The high speed of the craft makes it particularly important for collision avoidance alterations to be made early and positively, so that other vessels are aware, both visually and on radar, that effective action has been taken. Alterations should therefore be large enough to easily be detected, and made early enough to give adequate distance at the closest point of approach. In open waters the closest point of approach should be 1 mile ahead or 5 cables abeam or astern of the other vessel.

Care should be taken in the use of VHF for collision avoidance, especially at night or in poor visibility or when the identity of the other vessel cannot be guaranteed.

The International Regulations for Preventing Collisions at Sea must be strictly complied with at all times.

Radars must be monitored continuously and checked frequently one against the other by visual means. The data read from the ARPA radar should be treated with caution and not implicitly relied upon.

Neither vessel had in place any further detailed instructions or guidance regarding what actually constitutes a safe speed in restricted visibility. Additionally, in relation to the operating speed of both vessels, an assessment of the risks involved of high speed in restricted visibility had not been carried out.

1.12 CHANNEL NAVIGATION INFORMATION SERVICE (CNIS)

The IMO's resolution A.578 (14) defines Vessel Traffic Services (VTS) as:

Any service implemented by a competent authority designed to improve safety and efficiency of vessel traffic and the protection of the environment. The service shall have the capability to interact with marine traffic and to respond to traffic situations developing in the VTS area.

The following are extracts from the IMO resolution A.857 (20) Guidelines for VTS:

.9.1 An information service is a service to ensure that essential information becomes available in time for on-board navigational decision-making.

.9.2 A navigational assistance service is a service to assist on-board navigational decision-making and to monitor its effects.

.9.3 A traffic organisation service is a service to prevent the development of dangerous maritime traffic situations and to provide for safe and efficient movement of vessel traffic within the VTS area.

2.1 The purpose of VTS is to improve the safety and efficiency of navigation, safety of life at sea and the protection of the marine environment and/or the adjacent shore area, worksites and offshore installations from possible adverse effects of maritime traffic.

2.1.2 The type and level of service or services rendered could differ between both types of VTS; in a port or harbour VTS a navigational assistance service and/or a traffic organisation service is usually provided for, while in coastal VTS usually only an information service is rendered.

2.3.4 *When the VTS is authorised to issue instructions to vessels, these instructions should be result-orientated only, leaving the details of the execution, such as course to be steered or engine manoeuvres to be executed, to the master or pilot on board the vessel. Care should be taken that VTS operations do not encroach upon the master's responsibility for safe navigation or disturb the traditional relationship between master and pilot.*

As suggested in 2.1.2 above, there are two types of VTS: port/harbour and coastal, which can be found throughout the world. The former is a service provided for ships entering and leaving the confines of a port, and/or transiting within harbour limits, and the latter is concerned with traffic passing through an area outside harbour limits.

When entering the area covered by the Dover Strait TSS, all ships over 300gt report to Dover Coastguard, which deals with south-west bound traffic, or to Gris Nez Traffic (in France), which handles north-east bound traffic. The reporting system is mandatory, and the short title for the system is CALDOVEREP.

The following description is from the IMO's publication *Ship's Routing*:

The CNIS processing and display system receives inputs from the radar and VHF DF equipment, processes the information and presents it on any or all of six displays. Each display shows processed images (tracks) from any of the three radar inputs overlaid on a synthetic map of a selected area. New targets entering radar range are automatically tagged with a unique track number. The position course and speed information of up to 300 tracks is automatically updated and recorded, for each of the three radars, throughout the vessel's passage through the CNIS area, giving the CNIS a 900-track capability.

DOVER COASTGUARD maintain a continuous watch on traffic in the Dover Strait/Pas de Calais. Operators can add vessel information to the information processing and retrieval system database (such as name and cargo) and can display that supporting information on a separate screen. CNIS is capable of providing an automatic alarm to identify any track, which strays into an unauthorised area. VHF DF vectors appear when a VHF radio transmits on the frequency selected on the VHF DF equipment. Recording equipment automatically stores information from all tracks which can either be replayed on the system or specific track movements can be plotted onto an A0-size sheet of paper.

CNIS was introduced in 1972. It provides a 24-hour radio service for all shipping in the Dover Strait and is operated from the MRCC at Langdon Battery near Dover.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 ACTIONS TAKEN BY *DIAMANT*

2.2.1 Safe speed

It is appreciated that the criteria for determining a safe speed, especially aboard high-speed craft, are open for debate and, no doubt, can be interpreted in many different ways. However, a prudent approach, especially in an area such as the Dover Strait, could be to travel at no greater a speed than that in relation to the visibility and stopping distance of the vessel, having regard to the additional factors listed in Rule 6 (a).

Diamant, in accordance with her trial data, in an emergency situation, was able to come to a complete stop at her operating speed in 463 metres. Therefore, given the circumstances, it could be assumed that *Diamant's* speed at the time of the collision cannot be considered a safe speed in visibility of less than 463 metres. However, she was fitted with operational radar, which enabled the bridge team to detect other vessels beyond the visible range. What is unknown is how much further this was. A speed, greater than that in relation to the visibility and stopping distance of the vessel, can only be guaranteed a safe speed if there is an assured detection of all vessels at sufficient range so as to be able to avoid a collision in accordance with the Collision Regulations. Such assurance relies on those additional factors listed in Rule 6 (b), as well as the time required for detection, reaction and implementation of evasive action in the prevailing circumstances with an adequate safety margin to cater for the unexpected, and the heightened potential for damage to the vessel and injury to her crew and passengers in the event of an accident at high speed.

There can be no guarantee that *Diamant* would be able to detect all vessels at sufficient range so as to be able to avoid a collision in accordance with the Collision Regulations. *Diamant* was, therefore, travelling at a potentially unsafe speed.

A N Cockcroft and J N F Laeijer in their publication *A Guide to the Collision Avoidance Rules* state:

The word 'safe' is intended to be used in a relative sense. Every vessel is required to proceed at a speed which could reasonably be considered in the particular circumstances. If a ship is involved in a collision it does not necessarily follow that she was initially proceeding at an unsafe speed. In clear visibility, collisions can generally be attributed to bad look out, or to wrongful action subsequent to detection, rather than to a high initial speed.

They also state:

in restricted visibility a vessel making proper use of radar will normally be justified in going at a higher speed than that which would be acceptable for a vessel which does not have the equipment but not usually at the speed which would be considered safe for good visibility.

The master's decision to proceed at 29 knots in such conditions was borne out of an apparent general high-speed craft philosophy that, because of their manoeuvrability, a high speed can be a safe speed. However, it would be dangerous to assume this in all scenarios.

Recognising that there is always a possibility that small vessels and other floating objects might not be detected by radar at an adequate range (Rule 6(b)(iv)), it follows that a speed which relies on radar for detecting vessels at a sufficient range so as to be able to avoid collision, in accordance with the Collision Regulations, should not be regarded as a safe speed.

However, the practicality of following the above criterion in conditions of severely restricted visibility is questionable (eg the need to maintain steerage in conditions of zero visibility). Additionally, the commercial viability of shipping would be in danger of being undermined if the criterion was strictly applied, particularly in areas prone to restricted visibility.

In view of the above, a more pragmatic approach might be appropriate, such that a degree of reliance on radar for detection might be acceptable following a reasoned assessment of the risks in doing so.

2.2.2 Initial situation

When the bridge team on board *Diamant* first became aware that they would be involved in a passing situation with *Northern Merchant*, in this case resulting in a CPA of 3 cables, they either did not recognise it as a developing close quarters situation, or, if they did, not one that involved a risk of collision. The fact that no action was taken in accordance with Rule 19(d) supports this. In effect, they were using their clear visibility criterion for determining a close quarters situation and /or risk of collision. They did not recognise the need for an increased CPA in restricted visibility. Had they determined it to be a close quarters situation and/or risk of collision, they should have acted in accordance with Rule 19(d), which could, having passed South Goodwin light vessel, have been an alteration of course to starboard towards the amply available sea-room to the north. In addition, they also had the option of slowing down. However, as they did not consider that Rule 19(d) applied, they decided to maintain course and speed. This raises the issue of what constitutes a close quarters situation.

A N Cockcroft and J N F Laeijer state:

The distance at which a close quarters situation first applies has not been defined in miles, and is not likely to be, as it will depend upon a number of factors. The 1972 Conference (IMO Revision of the Collision Regulations) considered the possibility of specifying the distance at which it would be begin to apply but after lengthy discussions it was decided that this distance could not be quantified.

On the other hand, the Seafarers International Research Centre (SIRC), in a paper dealing with near miss encounters in the Dover Strait (Belcher P (2002) “Overtaking in the Dover Strait, an analysis of near miss encounters”) states:

From a review of literature on ships’ domains (Fuji and Tanka, 1971, Goodwin, 1977, Coldwell, 1983, Zhao, et al., 1993) it has been found that the domain required for a ship in congested waters can be approximated to a circular space with a radius of 8 cables. They also state: It might be argued that a criteria of a minimum passing distance of 8 cables is too stringent a measure for such a busy area. However, a passing distance of 3 cables or less, is on anyone’s measure, a very dangerous occurrence that could lead to a collision with only a very slight change in circumstances.

The speed at which a vessel is travelling will have a direct influence on the range at which a close quarters situation exists. The faster the speed, the more reaction and implementation distance will be required.

Why was an experienced high-speed craft master content with a CPA of 3 cables in visibility of 50 to 150 metres in such a busy seaway as the Dover Strait?

Crossing the Dover Strait several times a day on the same route, passing the same vessels in approximately the same positions, thereby anticipating their movements led to a certain degree of complacency. Margins of 3 cables became an accepted practice in clear visibility and, then, in all conditions of visibility.

The bridge team’s assumption that it was a green-green situation was incorrect, as *Northern Merchant* continued to show her port side throughout the accident (**Figure 7**). Their assumption was based on their anticipation of *Northern Merchant*’s likely passage. In not monitoring the ARPA course and speed display following their initial appraisal, contrary to Rule 7(b), they failed to maintain a full appraisal of the situation, contrary to Rule 5 of the Collision Regulations. Instead, reliance was placed solely on her CPA; the master neither requested, nor was offered, the ARPA’s course and speed reading.

Being under the impression that the situation was green-green, it seems the master anticipated that *Northern Merchant* would stand-on, for the following reasons: a perceived “unwritten rule” that high-speed craft will keep clear of all others, a general view by regular users of the Dover Strait that a CPA of 3 cables was acceptable and, given that acceptance, an alteration of course by *Northern Merchant* was unlikely.

2.2.3 Later situation

Having committed himself to a CPA of 2 to 3 cables, when *Northern Merchant*'s echo began to arc, because of side-lobing, as her aspect began to change when she altered course to starboard, *Diamant*'s master, mainly because of his green-green interpretation, always assumed the danger to be on his own starboard side. As a consequence, he altered course to port, initially by a few degrees only, in an attempt to regain confirmation of the situation and, in turn, increase the CPA and reduce the side-lobing on the radar display. He never considered an alteration of course to starboard. He assumed this would present *Diamant*'s port side to the approaching *Northern Merchant*.

At that time, there was a doubt as to whether a close quarters situation was developing. A prudent course of action then would have been to apply Rule 2(a): assume a close quarters situation and/or risk of collision and take all way off the vessel in accordance with Rule 19(e).

Only when *Northern Merchant*'s port side began to loom out of the fog, did *Diamant*'s master consider any emergency evasive action. Until that point, he had always been under the impression that both vessels would pass green-green. However, an alteration of course to starboard was no longer an option as it probably would have resulted in a more severe collision. *Diamant* was already slowly altering course to port, and it would have been much easier to manoeuvre the vessel into an emergency port turn.

Given the options available at the point when *Northern Merchant* appeared out of the fog, *Diamant*'s master's decision to continue altering course to port in an emergency turn probably reduced the severity of the collision compared with an alteration of course to starboard.

2.2.4 Situational appraisal

A crucial factor in the bridge team's assumption that *Diamant* would pass *Northern Merchant* green-green was their reliance on the CPA only. The master neither requested, nor was offered, the ARPA's course and speed reading. Had he either obtained, or been provided with, this information, he probably would have realised that both vessels continued to be involved in a crossing situation. This, of course, assumes that the course and speed readings obtainable from the radar were accurate.

2.2.6 Company procedures

A *Route Operating Manual*, which detailed company procedures and instructions in accordance with ISM Code requirements, was on board *Diamant*. Under collision avoidance, it stated that action to be taken to avoid a collision should be early, positive and large enough to be easily detected. It also made reference to CPAs ideally being 1 mile ahead and 5 cables abeam, albeit, in open waters. These distances were derived for the purpose of avoiding the need to comply with Sections II and III of the Steering and Sailing Rules. However, if those sections did become applicable, because of traffic density and other circumstances, then they were to be followed.

With regard to radar, it advised risk of collision should be frequently checked by visual means, and warned that the data read from ARPA should be treated with caution and not be relied upon implicitly.

Dover Strait is far from open waters. Therefore, the company should have in place procedures for coastal waters, with particular regard as to what constitutes a close quarters situation and/or risk of collision, and whether its vessels should keep clear of all other vessels (ie 1 mile ahead and 5 cables abeam) even if the Steering and Sailing Rules apply and there is no risk of collision.

2.2.7 Action when other vessel (*Northern Merchant*) became visible

When *Northern Merchant* became visible out of the fog, *Diamant's* master made an emergency manoeuvre to port, using what he thought was assistance from astern propulsion on the port engines. However, the action taken in principle was incorrect because the port buckets were to port with the port engines astern, and therefore, acted against the alteration of course to port.

Other emergency options were available as detailed by the Sindel model tests carried out by Warsash Maritime Centre. The six simulations carried out, with varying degrees of helm and engine movements, showed the minimum amount of advance and transfer achievable was insufficient to prevent a collision at a distance of 150 metres; the maximum reported visibility at the time. To prevent a collision, any emergency action would have to have been taken at a distance greater than 400 metres.

The model tests showed that the most effective action, given available sea room, was to use full power on both engines and maximum helm. Having said that, there was very little difference between the modelled manoeuvres.

2.2.8 Action after the collision

Passengers' reactions as to how the accident was handled were somewhat critical. The one factor about which they were all agreed, was that they felt they were not kept sufficiently informed as to what was happening. This led to uncertainty and confusion. It was also felt, generally, that the cabin staff lacked any knowledge regarding the situation.

The common factor in such accidents is that those on the bridge are very often heavily occupied in handling post-accident events. In such circumstances, it is necessary to prioritise what is required, and it may be that keeping the passengers informed is not necessarily considered to be of the highest priority at the time. It is also possible that those on the bridge might themselves be suffering from various states of shock. In addition, post-accident stress can result in those on the bridge temporarily losing their concept of the passage of time. What they believe is an interval of only 1 or 2 minutes may in fact be 5 to 10 minutes. It is not, therefore, a case that those on the bridge are unaware of the need to inform people of what is happening. The person best placed to make the initial calm and authoritative broadcast is very often someone not directly involved in handling the accident or its immediate aftermath. The potential difficulty with this, however, is that such a person may not be fully familiar with all the necessary facts to enable him/her to make such an announcement. It would mean that those on the bridge would have to divert precious time to brief someone.

Nonetheless, management should give thought to how this requirement can be met.

A number of people, both passengers and crew were severely shaken. Everyone wanted to know what had happened, whether they were safe, and what was going to happen next. In the aftermath of this accident, such information was, at best, patchy.

Providing regular and accurate information in an authoritative and calm manner is among the most important of all requirements in any passenger-carrying vessel involved in an emergency. The need is extremely well known but, as this accident demonstrates, it is often overlooked. The difficulties are recognised, especially when the communication channels are likely to be clogged. It does, however, need to be addressed by management.

2.3 ACTION TAKEN BY *NORTHERN MERCHANT*

2.3.1 Initial and later situations

As with *Diamant*, *Northern Merchant* was also travelling at a potentially unsafe speed. However, the master rarely reduced speed in restricted visibility.

Applying the same reasoning to *Northern Merchant's* speed, as to that of *Diamant*, in relation to stopping distances, it would have taken *Northern Merchant*, being a conventional craft, a greater distance than *Diamant* to come to an emergency stop.

Again, the master's decision to travel at full speed (20 to 21 knots) was based on normal operating practice. A reduction in speed was not considered, apart from allowing *Calais* more sea room to enter Dover harbour. After *Calais* was clear, the master resumed full speed. The visibility was never a factor.

When the bridge team on board *Northern Merchant* first became aware that they would be involved in a close quarters situation with *Diamant*, they should have acted in accordance with Rule 19(d) of the Collision Regulations.

It is accepted that, eventually, the master did alter course to starboard. However, the initial alteration of course was only by 7° to 10°. As an alteration of course, under the circumstances, it was inadequate and in contravention of Rule 8 of the Collision Regulations. Only when the master realised that *Diamant* was taking no avoiding action did he increase the alteration by ordering 20° starboard helm. However, this alteration, apart from not being readily apparent, was far too late.

The question which will be asked of *Northern Merchant's* action is: Why did the master initially only alter course by 7° to 10°, virtually standing on, with a CPA of 2 cables?

Being aware that the other vessel was a high-speed craft the master, in keeping with the perceived "unwritten rule", fully expected *Diamant* to keep clear and, once past South Goodwin light vessel, to alter course to starboard. When this did not happen, only then did he consider taking substantial avoiding action. Unfortunately, the application of only 20° starboard helm was insufficient to avoid a collision.

2.3.2 Company procedures

Northern Merchant carried a *Ship Operating Manual*, which detailed company procedures and instructions in accordance with ISM Code requirements. In it, under *prevention of collision*, it stated the need to review Rules 2, 6 and 19 of the *Collision Regulations* and that these Rules took precedence over commercial pressure and the requirements of the vessel.

Under fog and restricted visibility, it stated that main engines must be slowed and precautions carried out in accordance with the ICS Bridge Procedures Guide.

However, rather than quote adherence to the Collision Regulations, it would be far more beneficial to masters and watchkeepers if procedures and guidance were in place as to what constitutes compliance with the regulations, in particular with regard to close quarters situations and safe speed.

SECTION 3 - CONCLUSIONS

3.1 CAUSES AND CONTRIBUTING FACTORS

1. *Diamant's* potentially unsafe speed. [2.2.1]
2. An apparent general high-speed craft philosophy that, because of their manoeuvrability, a high speed can be a safe speed in all scenarios. [2.2.1]
3. A failure of *Diamant's* bridge team to recognise 2 to 3 cables as a close quarters situation. [2.2.2, 2.2.3]
4. A green-green passing assumption made by *Diamant's* bridge team. [2.2.2]
5. Complacency of *Diamant's* bridge team in its acceptance of small CPAs with other vessels. [2.2.2]
6. A failure of *Diamant's* bridge team to make continued use of the ARPA's course and speed display. [2.2.2]
7. An expectation, by the master of *Diamant*, that *Northern Merchant* would stand-on. [2.2.2]
8. A perceived "unwritten rule" that high-speed craft will keep clear of all other craft. [2.2.2]
9. Arcing of *Northern Merchant's* echo because of side-lobing. [2.2.3]
10. *Diamant's* master's decision to alter course to port. [2.2.3]
11. A failure by *Diamant's* bridge team to act in accordance with Rules 2(a) and 19(e). [2.2.3]
12. The lack of an effective risk assessment and instructions on board *Diamant* as to what constituted a close quarters situation and a safe speed in coastal waters. [2.2.6]
13. *Northern Merchant's* potentially unsafe speed. [2.3.1]
14. *Northern Merchant's* master's practice of rarely reducing speed in restricted visibility. [2.3.1]
15. A decision of *Northern Merchant's* master to alter course to starboard initially by only 7° to 10°. [2.3.1]
16. *Northern Merchant's* master's subsequent decision to alter course by ordering only 20° starboard helm. [2.3.1]

17. *Northern Merchant's* master's expectancy that *Diamant* would keep clear and alter course to starboard, once clear of the South Goodwin light vessel. [2.3.1]
18. A lack of company procedures and guidance on board *Northern Merchant* as to what constituted a close quarters situation and safe speed in coastal waters. [2.3.2]

3.2 OTHER FINDINGS

1. Under certain conditions it is possible that small displayed ARPA CPAs, could be zero. [2.2.4]
2. The most effective emergency action for *Diamant* to take was the use of full power on the engines and maximum helm. [2.2.7]
3. Providing regular and accurate information in a calm and authoritative manner is among the most important requirements in any passenger-carrying vessel involved in an emergency. [2.2.8]
4. A risk assessment, which identifies effective control measures for the operation of high-speed craft in restricted visibility, should enhance their safe operation. [2.4]
5. Other vessels which acknowledge the perceived "unwritten rule", develop an expectation that high-speed craft will keep clear in all instances, and, in doing so, where the Steering and Sailing Rules do come into force, fail to take action in compliance with them. [2.5.1]
6. A change to the Collision Regulations to specifically accommodate high-speed craft is not considered necessary by the MAIB. [2.5.1]
7. It is not the role of CNIS to intervene to prevent collisions. [2.6]
8. Had both vessels been fitted with VDR/AIS, a more thorough investigation could have been undertaken, especially in relation to positions, courses, speeds and bridge team actions. [2.7]
9. The subsequent action taken by both V Ships and Sea Containers Ltd should contribute to reducing the possibility of a similar accident occurring. [2.8]